Week 13

Keel Bone Chicken Project

03/30/26

Purdue University College of Engineering

This Week

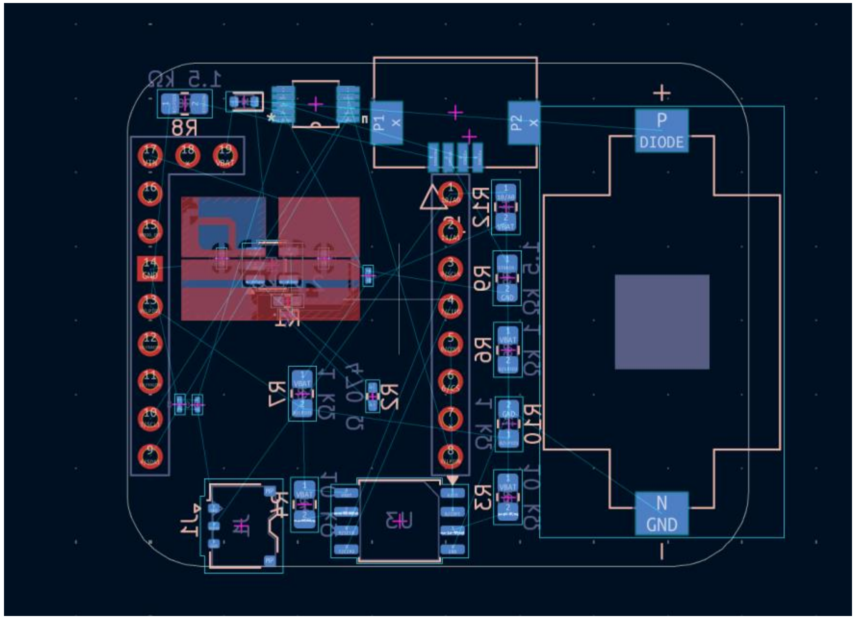
Weekly Progress

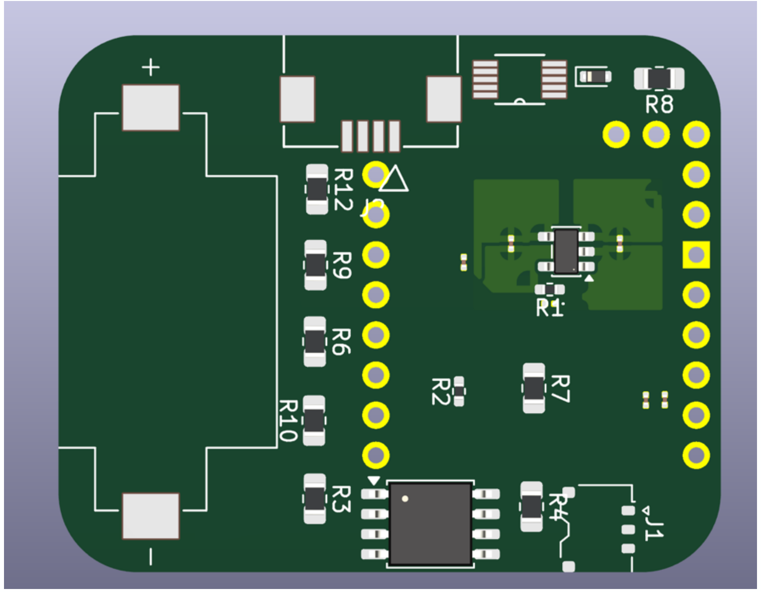
- PCB Updates
  - New sizing for PCB discovered
- Sensor Updates
  - Strain & force gauges received
  - Begin process of calibration on breadboard
- Data Logger Updates
  - Agriculture issues resolved
  - Device protocol changed
- Powercast Updates
  - NDA should be addressed to Professor Nawrocki soon
- Next Week

PCB Updates

New sizing for PCB discovered

- Remeasured backpack:
  - Exterior: 37 mm x 44 mm x 16.5 mm
  - Interior: 42 mm x 34 mm x 14 mm
- Changed sizing on second PCB variation:
  - Lost a bit of area





Sensor Updates

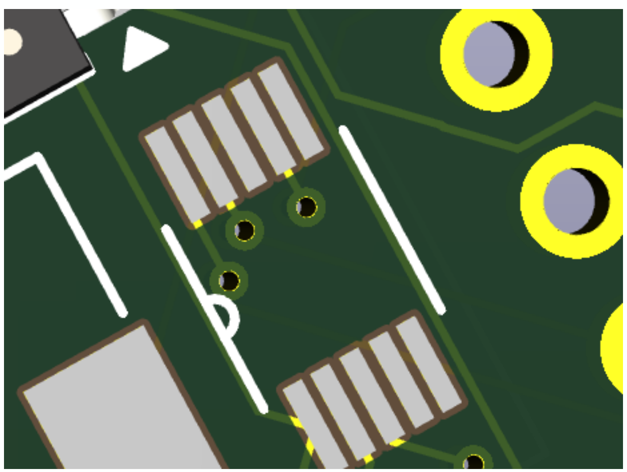
Strain & Force Gauges Received

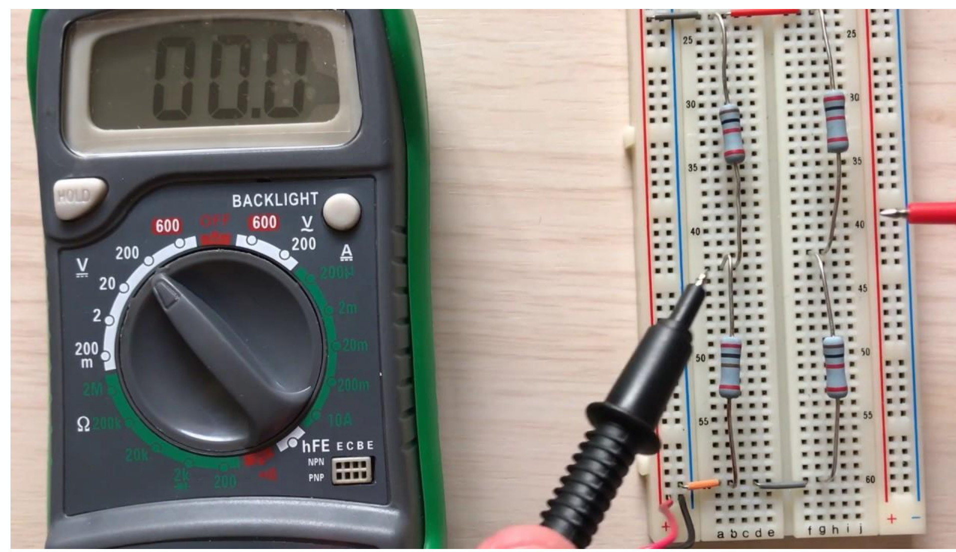
- Checked shipment areas for sensors
- Both strain and force have arrived

Sensor Updates

Begin process of calibration on breadboard

- Wheatstone bridge to calibrate strain gauge
  - ~3% tolerance, hopefully can be hit
- Test charge amplifier on PCB to calibrate force gauge





Data Logger Updates

Agriculture Issues Resolved

- Agriculture was using broken devices for some of the testing, got that resolved
- Needed a better fix for the excess power that might have been draining
- Set maximum records to an even higher constraint

Data Logger Updates

Device Protocol Changes

- Turned off unnecessary parts of the device if not going to be used
- Can be easily changed or modified

```cpp
#include "Nicla_System.h"
#include "Arduino_BHY2.h"
#include <ArduinoBLE.h>

#define BLE_SENSE_UUID(val) ("19b10000-" val "-537e-4f6c-d104768a1214")

const int VERSION = 0x00000000;

BLEService service(BLE_SENSE_UUID("0000"));

BLEUnsignedIntCharacteristic versionCharacteristic(BLE_SENSE_UUID("1001"), BLERead);
BLEFloatCharacteristic temperatureCharacteristic(BLE_SENSE_UUID("2001"), BLERead);
// BLEUnsignedIntCharacteristic humidityCharacteristic(BLE_SENSE_UUID("3001"), BLERead);
BLEFloatCharacteristic pressureCharacteristic(BLE_SENSE_UUID("4001"), BLERead);

BLECharacteristic accelerometerCharacteristic(BLE_SENSE_UUID("5001"), BLERead | BLENotify, 3 * sizeof(float));
BLECharacteristic gyroscopeCharacteristic(BLE_SENSE_UUID("6001"), BLERead | BLENotify, 3 * sizeof(float));
BLECharacteristic quaternionCharacteristic(BLE_SENSE_UUID("7001"), BLERead | BLENotify, 4 * sizeof(float));

BLECharacteristic rgbLedCharacteristic(BLE_SENSE_UUID("8001"), BLERead | BLEWrite, 3 * sizeof(byte));

// BLEFloatCharacteristic bsecCharacteristic(BLE_SENSE_UUID("9001"), BLERead);
// BLEIntCharacteristic  co2Characteristic(BLE_SENSE_UUID("9002"), BLERead);
BLEUnsignedIntCharacteristic gasCharacteristic(BLE_SENSE_UUID("9003"), BLERead);

String name;

Sensor temperature(SENSOR_ID_TEMP);
```

```cpp
}

// Handlers remain the same as your original snippet...
void blePeripheralDisconnectHandler(BLEDevice central){
  nicla::leds.setColor(red);
}

// Unnecessary Characteristics
void onTemperatureCharacteristicRead(BLEDevice central, BLECharacteristic characteristic){
  temperatureCharacteristic.writeValue(temperature.value());
}

void onHumidityCharacteristicRead(BLEDevice central, BLECharacteristic characteristic){
  uint8_t humidityValue = (uint8_t)(humidity.value() + 0.5f);
  humidityCharacteristic.writeValue(humidityValue);
}

void onPressureCharacteristicRead(BLEDevice central, BLECharacteristic characteristic){
  pressureCharacteristic.writeValue(pressure.value());
}

void onBsecCharacteristicRead(BLEDevice central, BLECharacteristic characteristic){
  bsecCharacteristic.writeValue((float)bsec.iaq());
}

void onCo2CharacteristicRead(BLEDevice central, BLECharacteristic characteristic){
  co2Characteristic.writeValue((uint32_t)bsec.co2_eq());
}

void onGasCharacteristicRead(BLEDevice central, BLECharacteristic characteristic){
  gasCharacteristic.writeValue((unsigned int)gas.value());
}
//Stop Unncessecary Characteristics
```

Powercast Updates

NDA should be addressed to Professor Nawrocki soon

- Asked Dan Harrist for a version of the NDA to be sent to Professor Nawrocki
  - NDA should be arriving to purdue email soon
- Also asked for clarification on P2110B and P1110B

Next Week

Next Week's Plan

- Finalize Paper
- Ask Omosalewa for a recording of the data as it is collected
  - Match up chicken results to chicken video
  - Edit the video so such results are shown
- Do final data analysis on data collected

Thank You

Purdue Polytechnic Institute

PURDUE UNIVERSITY | College of Engineering